

W04 ENHANCED WATER QUALITY | O (MAX : 1 PT)

Intent : Provide access to drinking water without unpleasant taste, odor and appearance.

Summary : This WELL feature requires projects to provide drinking water that meets thresholds on chemicals that affect aesthetics and taste concerns.

Issue : Even when health-based thresholds for water quality are met, water may be found unappealing to drink because of taste, odor and appearance concerns. For example, high levels of chloride contribute to a salty taste and iron can give the water a reddish appearance.¹ Therefore, some regulatory bodies set non-enforceable limits based on human detectability and acceptability for these substances.^{2,3}

Solutions : Like pollutants with health-based concerns, the treatment system to address nuisance chemicals depends on the contaminant of interest. Treatment options include filtration with carbon media and reverse osmosis.

Impact : By managing nuisance chemicals, projects can provide more appealing and palatable drinking water.

PART 1 MEET THRESHOLDS FOR DRINKING WATER TASTE (MAX : 1 PT)

For All Spaces:

Water delivered to the project for human consumption meets the following thresholds:

- a. Aluminum ≤ 0.2 mg/L.²
- b. Chloride ≤ 250 mg/L.²
- c. Copper ≤ 1 mg/L.²
- d. Manganese ≤ 0.05 mg/L.
- e. Iron ≤ 0.3 mg/L.²
- f. Silver ≤ 0.1 mg/L.²
- g. Sodium ≤ 270 mg/L.³
- h. Sulfate ≤ 250 mg/L.²
- i. Zinc ≤ 5 mg/L.²
- j. Total Dissolved Solids (TDS) ≤ 500 mg/L.²
- k. Free Chlorine ≤ 1.25 mg/L.⁴

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. U.S. Environmental Protection Agency. National Secondary Drinking Water Regulations. <https://www.epa.gov/dwstandardsregulations/secondary-drinking-water-standards-guidance- nuisance-chemicals>. Published 2017. Accessed.
3. Health Canada. Guidelines for Canadian Drinking Water Quality—Summary Table. <https://www.canada.ca/en/health-canada/services/environmental-workplace-health/reports-publications/water-quality/guidelines-canadian-drinking-water-quality-summary-table.html>. Published 2017. Accessed.
4. Crider Y, Sultana S, Unicomb L, Davis J, Luby SP, Pickering AJ. Can you taste it? Taste detection and acceptability thresholds for chlorine residual in drinking water in Dhaka, Bangladesh. Science of The Total Environment. 2018;613-614:840-846.